

Arnav Meduri

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EDUCATION

Duke University

Expected: May 2028

Bachelor of Science in Computer Science and Statistics | GPA: 3.92

Durham, NC

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Databases, Applied Machine Learning, Artificial Intelligence, Statistical Computing, Linear Algebra, Regression Analysis, Probability Theory

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript/Typescript, HTML/CSS, SQL (PostgreSQL, MySQL), R, MATLAB, Assembly

Frameworks/Technologies: React/Next.js, Node.js, Angular, MongoDB, PyTorch, NumPy, Pandas, AWS, Linux, Unix

Skills: Object-Oriented Programming, RESTful APIs, AI/ML, Full Stack Development, Unit Testing, Git/GitHub

EXPERIENCE

Red Hat

May 2026 –

Software Engineer Intern

Raleigh, NC

Ecolytics (ESG Analytics Startup)

May 2025 – Jan 2026

Software Development Intern

Raleigh, NC (Remote)

- Developed a browser extension for AI environmental impact monitoring using JavaScript and HTML/CSS, enabling 40+ individuals and enterprise teams to offset 1,100+ kg of carbon dioxide-equivalent emissions.
- Built a React-based standards navigation tool with Fuse.js fuzzy search and dynamic filtering, helping clients efficiently navigate 480 B Corp certification requirements across 5 environmental impact areas.
- Implemented frontend performance optimizations (e.g., render-blocking resource elimination, unused CSS removal) on the company website, reducing average page load time by 1.8s and improving Lighthouse performance score from 42 to 78.

Authorship (Legal Tech Startup)

May 2025 – Aug 2025

Software Engineer Intern

Raleigh, NC (Remote)

- Built a React-based two-party case management system with multipart document uploads and DynamoDB-based data management, processing 80+ legal disputes across IP, compliance, and contract enforcement.
- Integrated LLaMA 3 8B via the Hugging Face Inference API to automate case resolution summarization; utilized AWS Lambda for backend inference requests and Amazon S3 for file storage.

Columbia University Irving Medical Center

June 2023 – April 2025

Machine Learning Research Intern

Durham, NC (Remote)

- Architected a ResUNet-based segmentation model for retinal tear detection in patients with acute symptomatic posterior vitreous detachments, achieving a 96% F1 score on clinician-annotated OCT B-scans.
- Developed image preprocessing pipelines (i.e., normalization, downsampling) with OpenCV and Torchvision to generate segmentation masks from 250+ B-scans for model training.
- Collaborated with vitreoretinal specialists to validate model outputs against expert-annotated ground truth labels.

PROJECTS

Minimal Cell Proteomics | *Biopython, EVcouplings, HMMER, PyMOL*

- Applied hidden Markov models (HMMER) to assign putative functions to 12 uncharacterized proteins in the JCVI-syn3A proteome based on PFAM domain homology.
- Leveraged pseudolikelihood-based maximum entropy models within the EVCouplings framework to identify 33 statistically significant inter-protein interactions across 110 PFAM-mapped protein families.

Pantry Patrol | *AppSheet REST API, Google Cloud Platform*

- Developed a cross-platform food pantry management application integrating AppSheet REST API for record management, Gemini API for automated item classification, and Google Cloud Firestore for backend data storage.
- Collaborated with the Food Bank of Central and Eastern NC and the UNC-Chapel Hill Gillings School of Public Health to deploy the application across 10+ pantries, reaching 3,000+ households.

Pacas Consumer Analytics | *Python (pandas, NumPy), R (dplyr, tidymodels)*

- Built a Python-based data pipeline to segment 80K retail transactions; used logistic regression to identify 6 high-volume products (~15% of catalog) associated with repurchase behavior.
- Performed purchase frequency modeling in R to quantify a 17% 60-day repurchase rate across customer segments.

ADDITIONAL INFORMATION

- **Research Awards and Conferences:** Regeneron Science Talent Search Scholar; NC Student Academy of Science Winner in Computational Biology; Presenter at ARVO 2024 and SPIE Photonics West 2025
- **Programming Awards:** 1st Place at Lenovo & NCBC's Ready, Set, App! Competition; 2x American Computer Science League National Finalist; Samsung Solve for Tomorrow State Finalist; eCYBERMISSION National Winner
- **Other Activities:** Technify (Director); Duke Applied Machine Learning Group (Software Engineer); Product Lab (Software Engineer); Duke Impact Investing Group (Consulting Analyst)